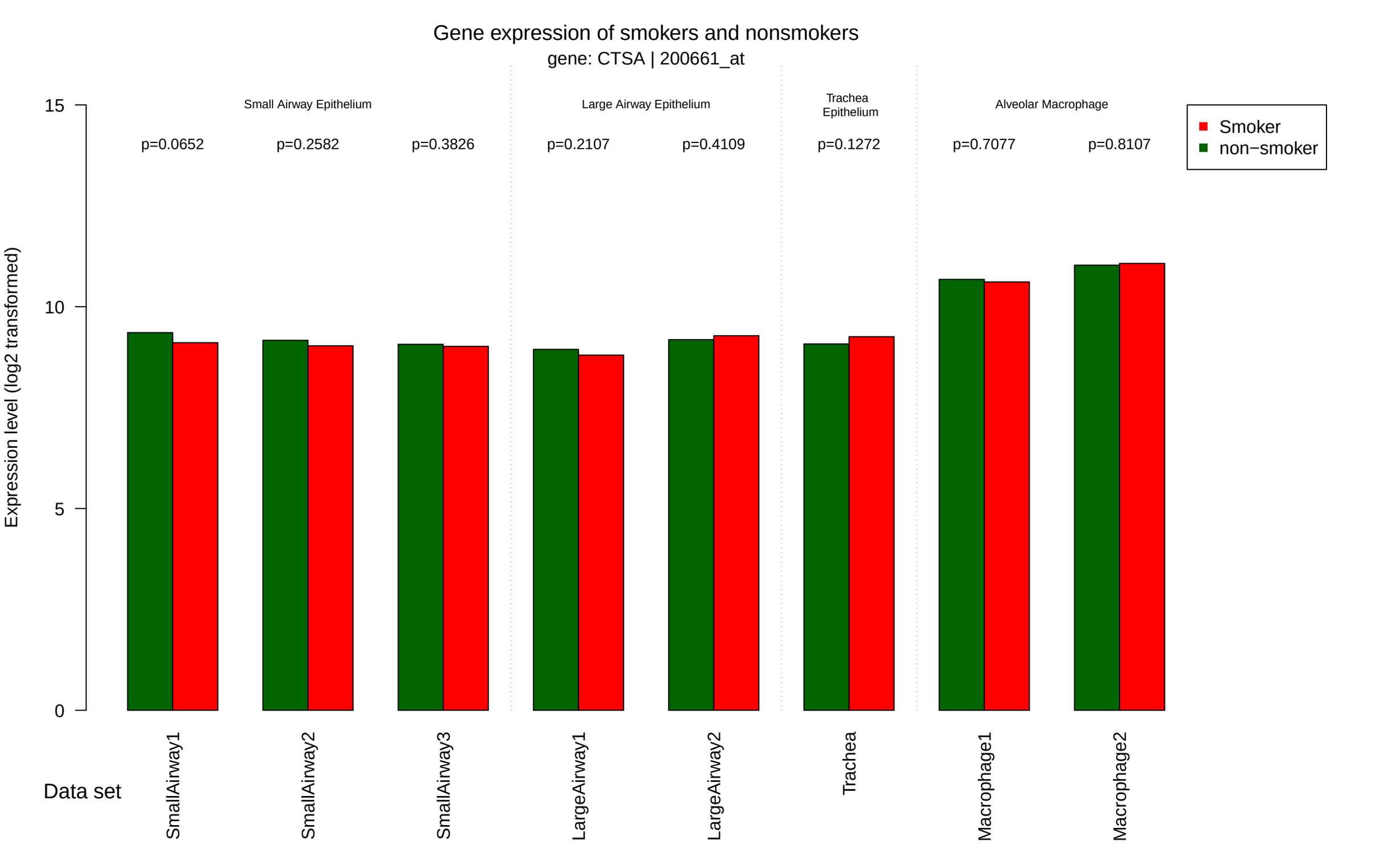
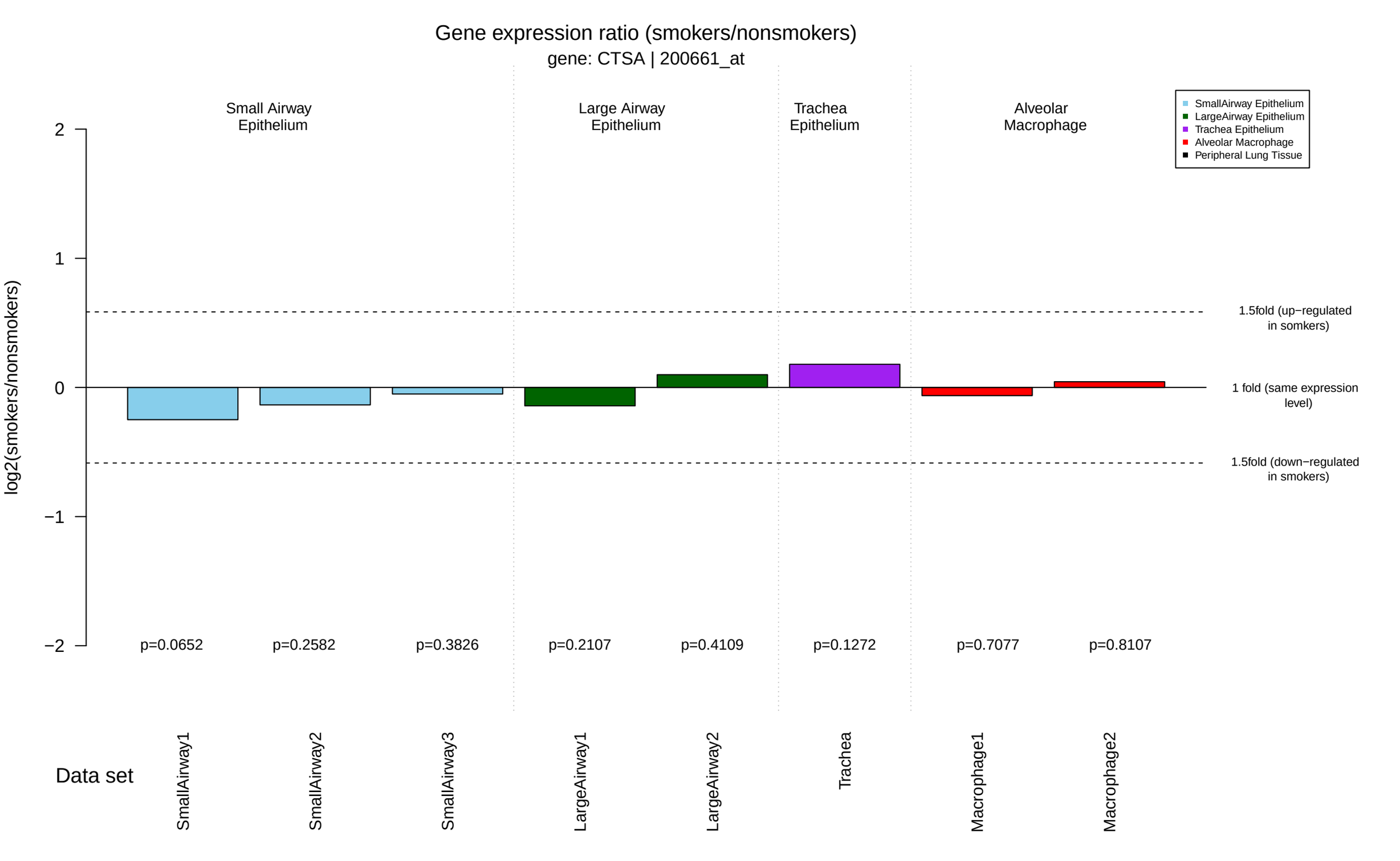
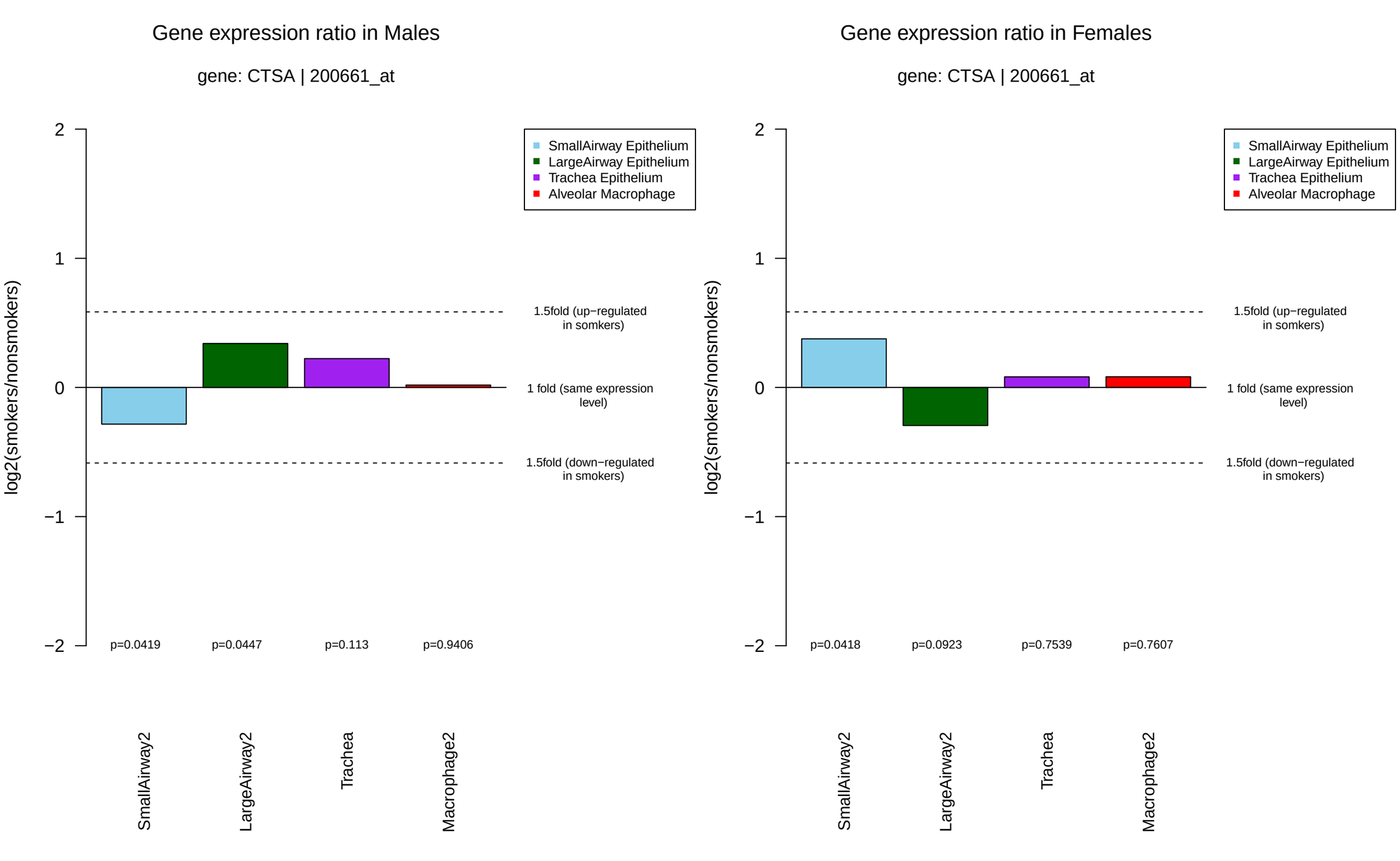
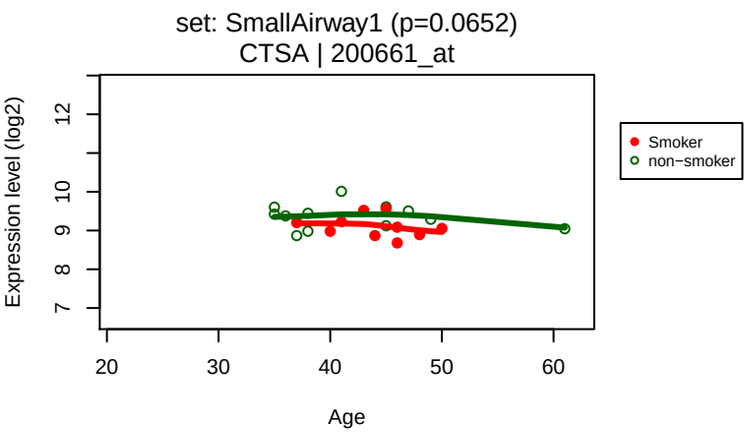
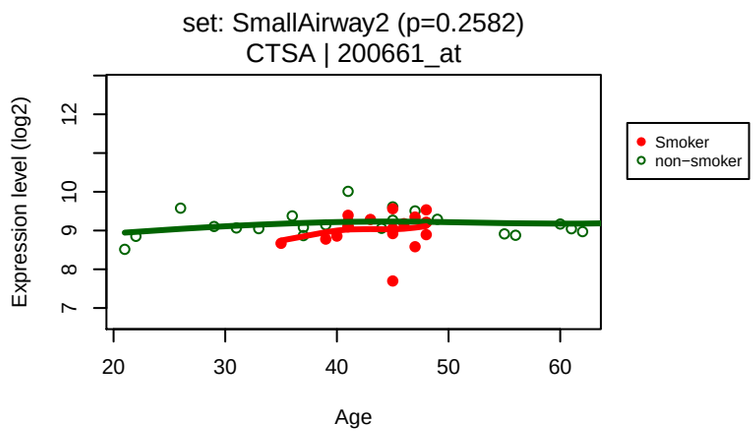
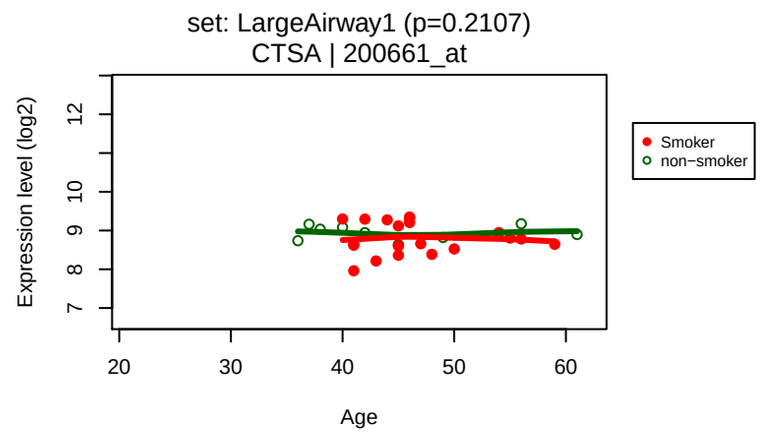

Gene expression of smokers and nonsmokers
gene: CTSA | 200661_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0652
p=0.2582
p=0.3826
p=0.2107
p=0.4109
p=0.1272
p=0.7077
p=0.8107
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

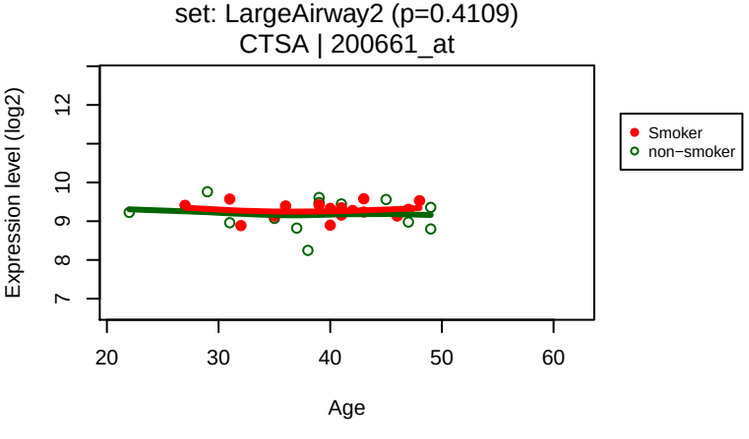

Gene expression ratio (smokers/nonsmokers)
gene: CTSA | 200661_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0652
p=0.2582
p=0.3826
p=0.2107
p=0.4109
p=0.1272
p=0.7077
p=0.8107
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

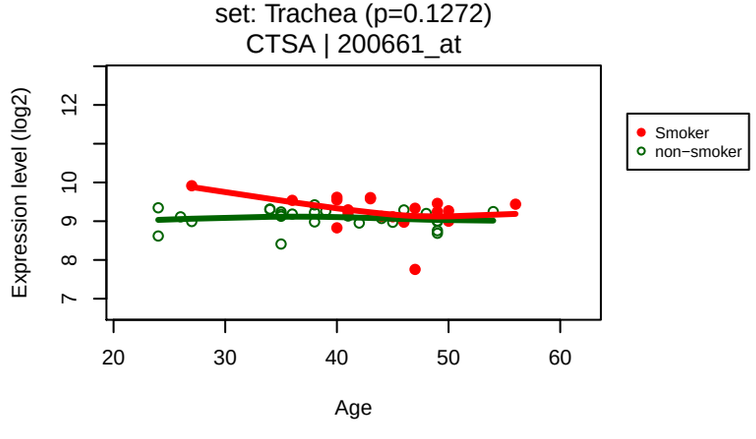

Gene expression ratio in Males
gene: CTSA | 200661_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0419
p=0.0447
p=0.113
p=0.9406
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CTSA | 200661_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0418
p=0.0923
p=0.7539
p=0.7607
SmallAirway2
LargeAirway2
Trachea
Macrophage2

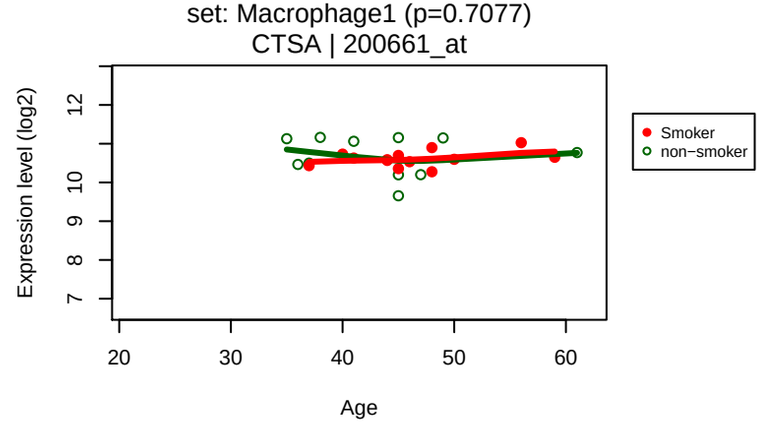

set: SmallAirway1 (p=0.0652)
CTSA | 200661_at
Expression level (log2)
Age
Smoker
non−smoker

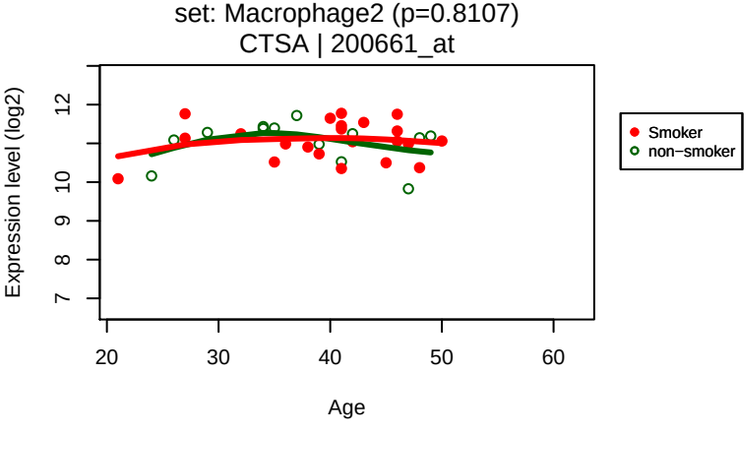

set: SmallAirway2 (p=0.2582)
CTSA | 200661_at
Expression level (log2)
Age
Smoker
non−smoker

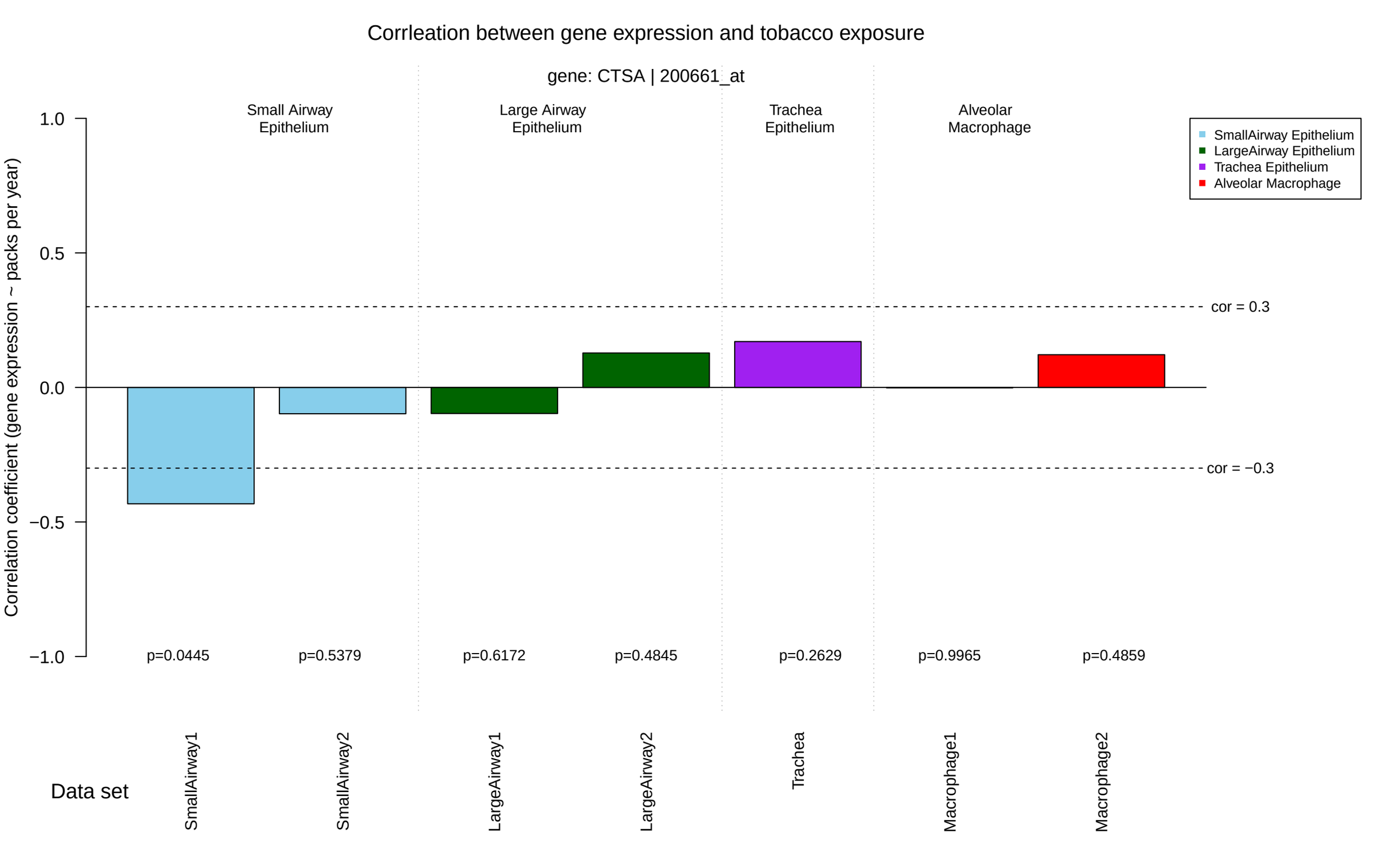

set: LargeAirway1 (p=0.2107)
CTSA | 200661_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4109)
CTSA | 200661_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1272)
CTSA | 200661_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7077)
CTSA | 200661_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8107)
CTSA | 200661_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CTSA | 200661_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0445
p=0.5379
p=0.6172
p=0.4845
p=0.2629
p=0.9965
p=0.4859
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2